

Python day05-lists

#multiple datatype assign

```
n_list=[2.9,3000,'Hello',"Welcome to python list ",  
True]  
print(n_list)
```

Nested List Example1

```
n_list = ["Happy", [2, 0, 1, 5]]
```

Nested indexing

```
print(n_list[0][3])  
print(n_list[1][2])
```

Nested indexing

```
print(my_list[1][1])  
print(my_list[2][2])
```

Negative indexing in lists

```
my_list = ['p','r','o','b','e']  
print(my_list[-1])
```

```
print(my_list[-5])
```

Appending and Extending lists in Python

```
odd = [1, 3, 5]
```

```
odd.append(7)
```

```
print(odd)
```

```
odd.extend([9, 11, 13])
```

```
print(odd)
```

Concatenating and repeating lists

```
odd = [1, 3, 5]
```

```
print(odd + [9, 7, 5])
```

```
print(["re"] * 3)
```

#USIN A WHILE LOOP

thislist = ["apple", "banana", "cherry"]

i = 0

while i < len(thislist):

 print(thislist[i])

 i = i + 1

#PYTHON SORTS LIST

thislist = ["orange", "mango", "kiwi", "pineapple",
"banana"]

thislist.sort()

print(thislist)

Output:['banana', 'kiwi', 'mango', 'orange', 'pineapple']

thislist = [100, 50, 65, 82, 23]

thislist.sort()

print(thislist)

#sort decending

thislist = ["orange", "mango", "kiwi", "pineapple",
"banana"]

thislist.sort(reverse = True)

print(thislist)

#python copy lists

thislist = ["apple", "banana", "cherry"]

mylist = thislist.copy()

print(mylist)

#python insert

fruits = ['apple', 'banana', 'cherry']

fruits.insert(1, "orange")

print(fruits)

#remove specified index

thislist = ["apple","banana","cherry"]

thislist.pop(1)

print(thislist)

#clear the results

Clear the list content:

thislist = ["apple","banana","cherry"]

thislist.clear()

print(thislist)

#python list count

fruits = [1, 4, 2, 9, 7, 8, 9, 3, 1]

x = fruits.count(9)

print(x)

Program to demonstrate

lst = [10, 20, 30, 40]

Create a copy of the list

new lst = lst.copy()

print("Original List :", lst)

print("Copied List :", new lst)

#list index() method

lst = [10, 20, 30, 40, 50]

value = 30

position = lst.index(value)

print("List :", lst)

print("Value :", value)

print("Index :", position)

#python pop

t = ['a', 'b', 'c']

x = t.pop(1)

print (t)

['a', 'c']

print (x)

b

#python remove

t = ['a', 'b', 'c']

t.remove('b')

print (t)

['a', 'c']

#python list built in functions

nums = [3, 41, 12, 9, 74, 15]

print len(nums)

#python spilit

s = 'pining for the fjords'

t = s.split()

print (t)

['pining', 'for', 'the', 'fjords']

print t[2]

python tuple operations

Creating a tuple

fruits = ("apple", "banana", "orange", "mango")

print("Tuple:", fruits)

Accessing elements

print("First fruit:", fruits[0])

print("Second fruit:", fruits[1])

print("Last fruit:", fruits[-1])

numbers = (10, 20, 30, 40, 50)

print("Tuple:", numbers)

print("Length:", len(numbers))

numbers = (10, 20, 30, 40, 50, 60)

print("Original:", numbers)

print("First 3:", numbers[:3])

print("Middle:", numbers[2:5])

print("Last 3:", numbers[-3:])

student = ("Arun", 20, "Python")

name, age, course = student

print("Name:", name)

print("Age:", age)

print("Course:", course)

numbers = (10, 20, 10, 30, 10, 40)

count()

print("10 occurs:", numbers.count(10), "times")

index()

print("Position of 30:", numbers.index(30))

fruits = ("apple", "banana", "orange", "mango")

students = ("Arun", "Bala", "Kumar", "Ravi")

print("Students:")

for student in students:

 print(student)

numbers = (10, 25, 5, 40, 15)

print("Numbers:", numbers)

print("Sum:", sum(numbers))

print("Maximum:", max(numbers))

print("Minimum:", min(numbers))

#compare two tuples

tuple1 = (1, 2, 3)

tuple2 = (1, 2, 4)

```
if tuple1 == tuple2:  
    print("Both tuples are equal")  
else:  
    print("Tuples are different")
```

```
#length  
tuple1 = (10, 20, 30, 40)
```

```
tuple2 = tuple1
```

```
print("Tuple 1:", tuple1)  
print("Tuple 2:", tuple2)
```

```
fruit = input("Enter a fruit: ")
```

```
if fruit in fruits:  
    print(fruit, "is present in the tuple")  
else:
```

print(fruit, "is not present in the tuple")

python dictionary operation

#creating a dictionary

student = {

"name": "Aushvanth",

"age": 20,

"department": "Civil Engineering"

}

print(student)

#accessing values

print(student["name"])

print(student["department"])

#adding new item

student["college"] = "Sudharsan Engineering College"

print(student)

#updating a value

student["age"] = 21

#removing a item

student.pop("age")

#using delete

del student["college"]

#important dictionary methods

student = {

 "name": "Aushvanth",

 "age": 20,

 "department": "Civil"

}

print(student.keys())

print(student.values())

print(student.items())

#using get()

print(student.get("name"))

print(student.get("marks"))

#key and values

for key, value in student.items():

 print(key, ":", value)

#nested dictionary

students = {

 "student1": {

 "name": "Aushvanth",

 "mark": 85

 },

 "student2": {

 "name": "Rahul",

 "mark": 90

 }

}

print(students["student1"]["name"])S